

Date: 5th April, 2026



ANRF AISEHack 2026

*Finale presentation /
Theme : 1 Flood
Detection(IBM)*

Jatin

Sara

Rida

Team Triverse

Your salient contributions

Modelling strategy

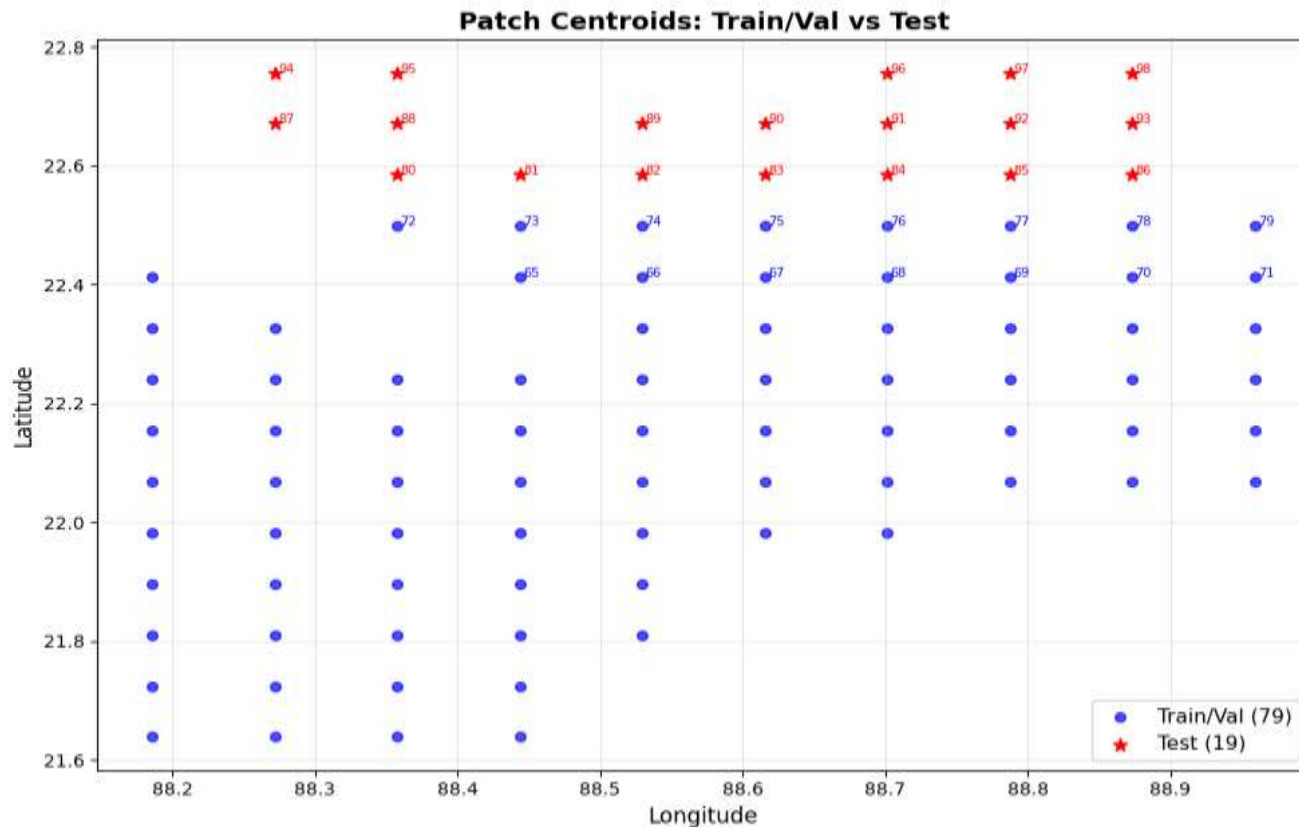
Our overall strategy is a multi-modal flood segmentation ensemble that combines a fine-tuned Prithvi geospatial foundation model with a MaxViT-UNet, using 18-band multi-temporal, hydrological, and topographic inputs to separate floodwater from permanent water bodies.

Training

In training, we fine-tune Prithvi and train MaxViT-UNet on labeled flood patches using multi-band normalized inputs. We use class-weighted segmentation loss, data augmentation, validation-based threshold tuning, and ensembling to improve flood IoU.

Results + Actions

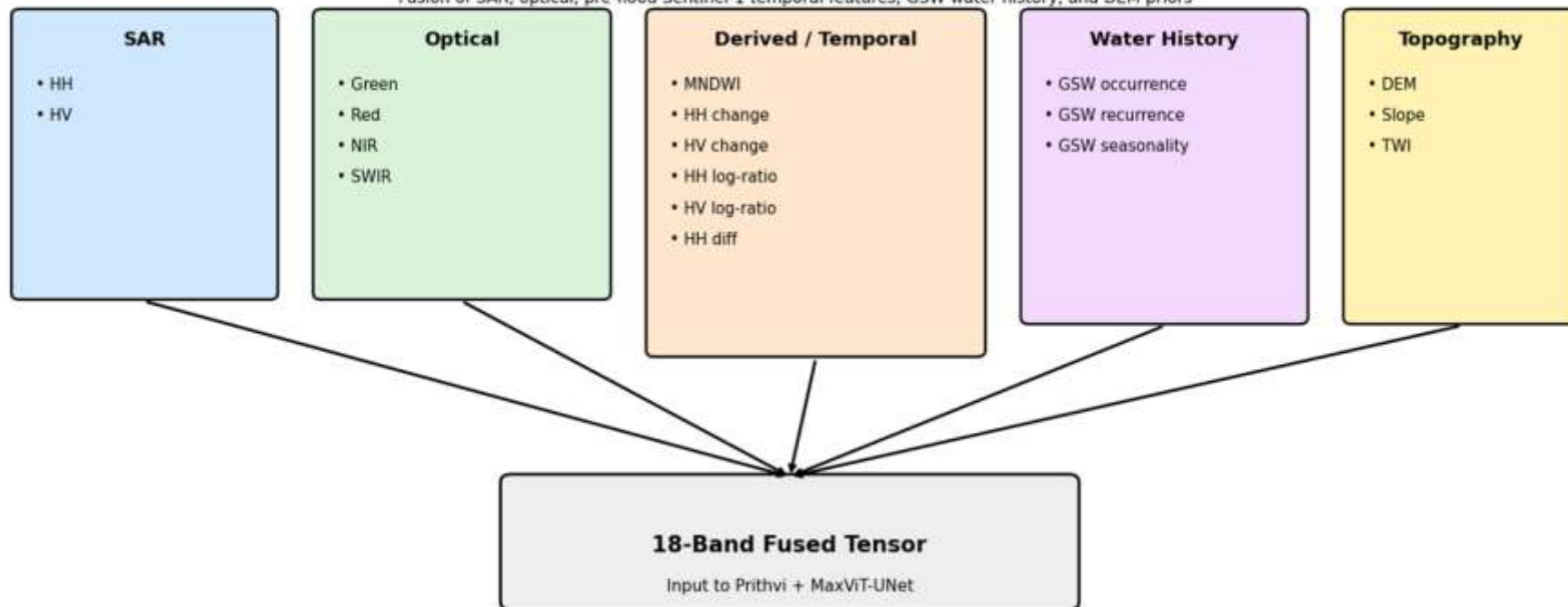
We fine-tuned Prithvi and trained MaxViT-UNet on multi-band flood patches with class-weighted loss and augmentation, then ensembled their outputs. This improved flood detection over baseline, with our best setup using threshold tuning and post-processing to raise leaderboard performance.



Spatial analysis reveals test patches (red) are geographically north of labeled data. Patches 70-79 (dark blue) are closest (~9.6 km), while 1-10 are >100 km away, validating our non-random validation split strategy.

18-Band Multi-Modal Feature Stack

Fusion of SAR, optical, pre-flood Sentinel-1 temporal features, GSW water history, and DEM priors



Goal: Improve flood vs permanent water discrimination using temporal, hydrological, and terrain context

Grouped 18-Band Visualization (excluding temporal bands 7-11)
20240529_EO4_RES2_fl_pid_001_image.tif

SAR: HH



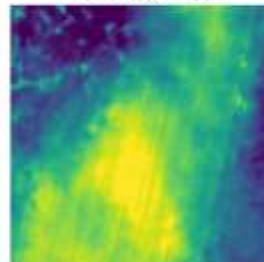
SAR: HV



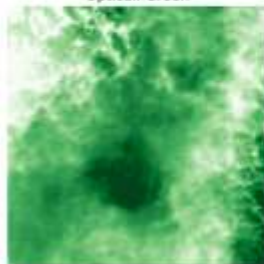
False Color (NIR-Red-Green)



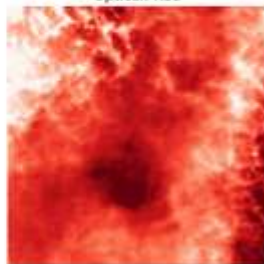
Water Index: MNDWI



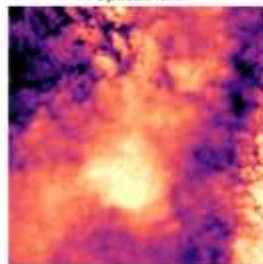
Optical: Green



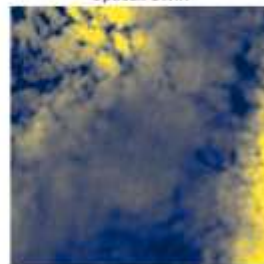
Optical: Red



Optical: NIR



Optical: SWIR



GSW: Occurrence



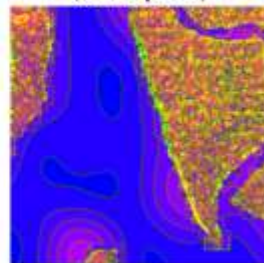
GSW: Recurrence



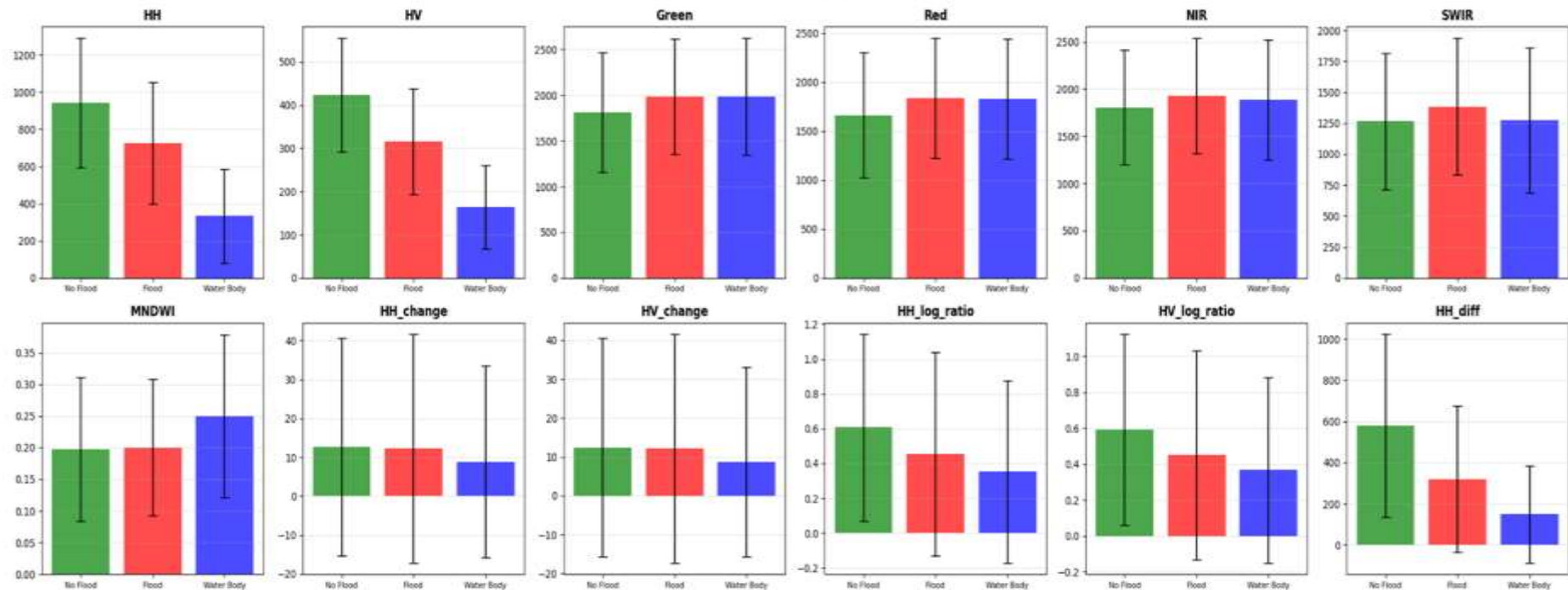
GSW: Seasonality



Topography Composite
(DEM-Slope-TWI)



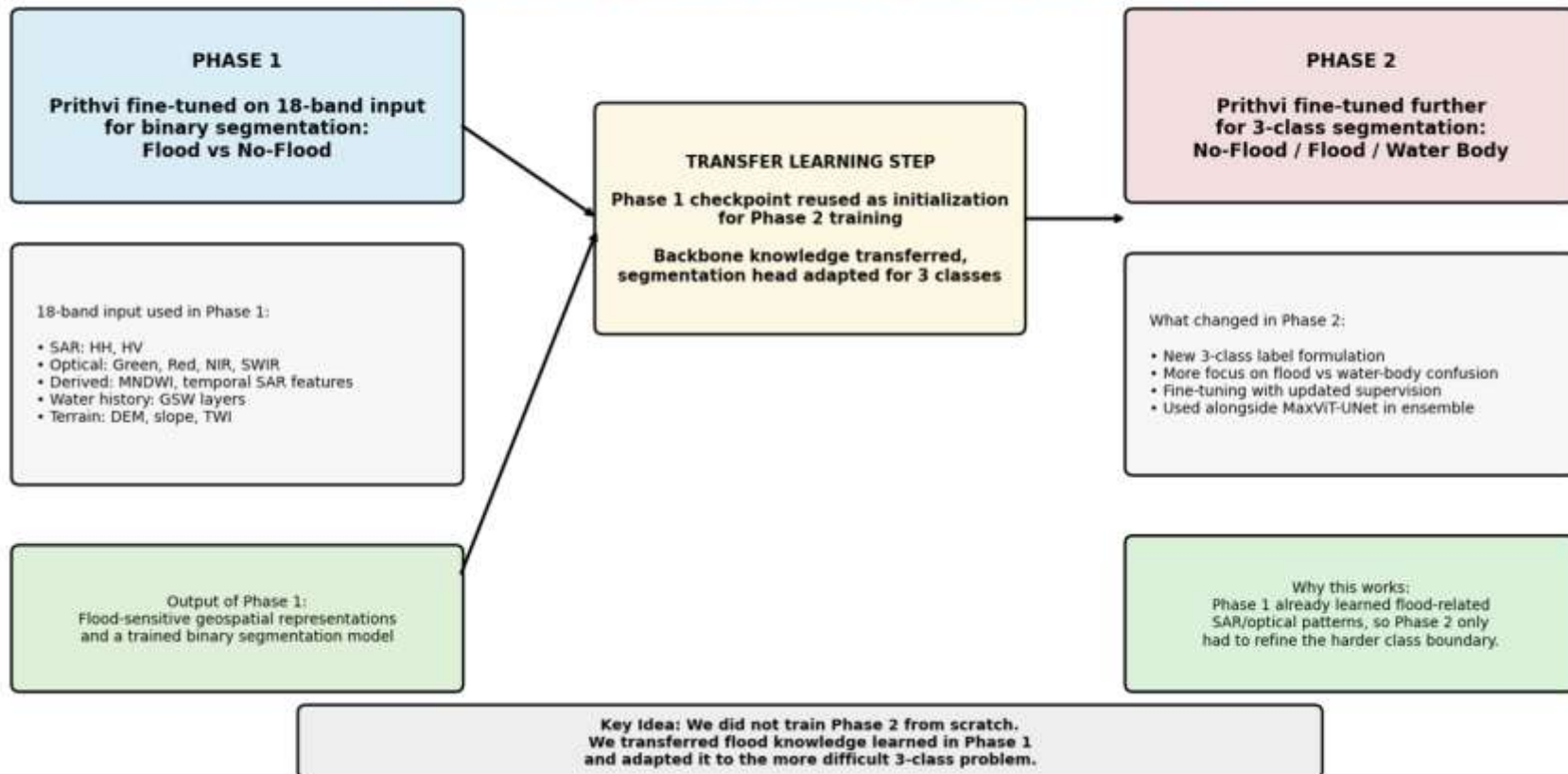
Band Values by Class (mean $\pm$ std)



Prithvi Transfer Learning: From Phase 1 to Phase 2

Using a geospatial foundation model trained on binary flood mapping and adapting it to a 3-class segmentation task.

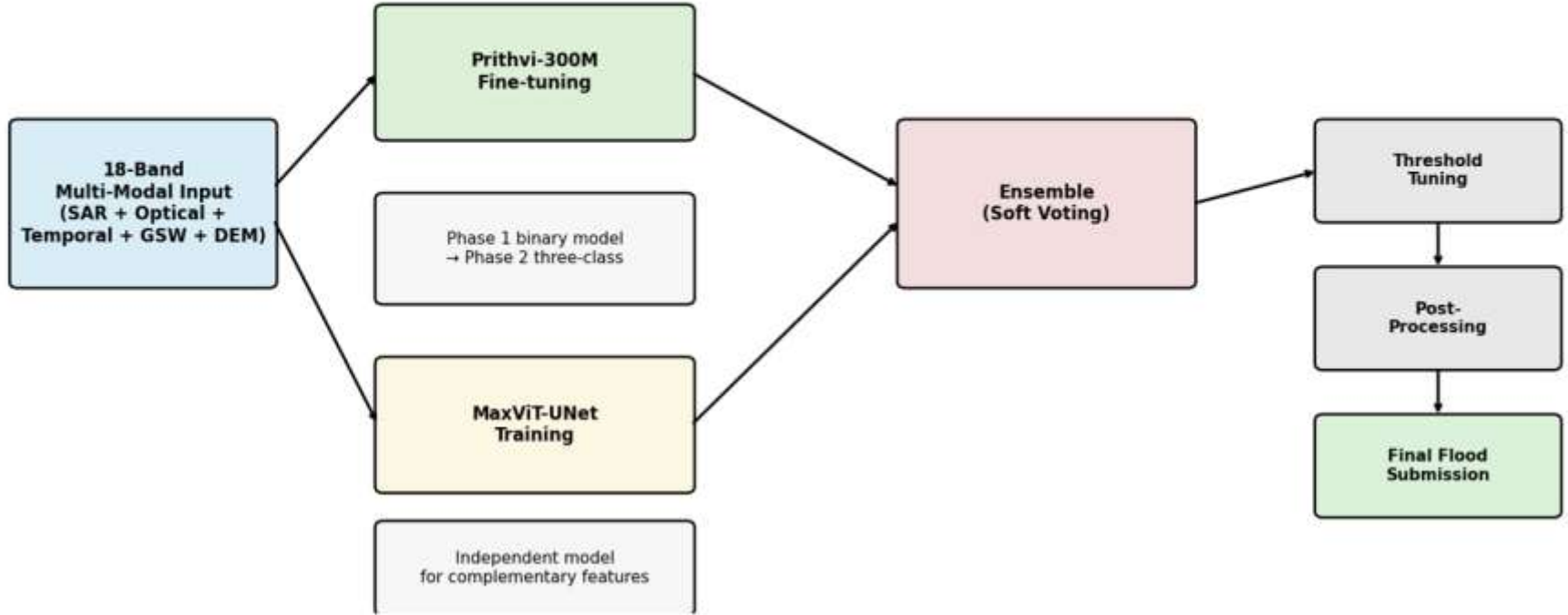
Prithvi backbone retained spatial-spectral flood representations; only task adaptation was required for Phase 2



Modeling Strategy: Prithvi + MaxViT Ensemble

Training: class-weighted loss + augmentation + validation on geographically closer patches

Goal: improve flood vs permanent water discrimination in Phase 2



Key Results – *Why, what and how of the workings?*

S1

What is working?

We used 18-band multi-modal data—EOS-4 SAR, optical bands, May 10 Sentinel-1 pre-flood SAR, temporal change features, GSW occurrence/recurrence/seasonality, and DEM/slope/TWI—to train fine-tuned Prithvi and MaxViT-UNet with class-weighted loss, augmentation, flood-rich oversampling, threshold tuning (~0.5), and ensemble inference.

● Implemented

All core setup criteria
done

S2

How is it working? (Quantitative evals)

It is working through improved flood IoU and leaderboard score over the baseline, with the biggest gains coming after adding May 10 Sentinel-1 temporal features, GSW/DEM context, class-weighted training, and ensemble + threshold tuning.

S3

Why is it working? (Reasoning and analysis)

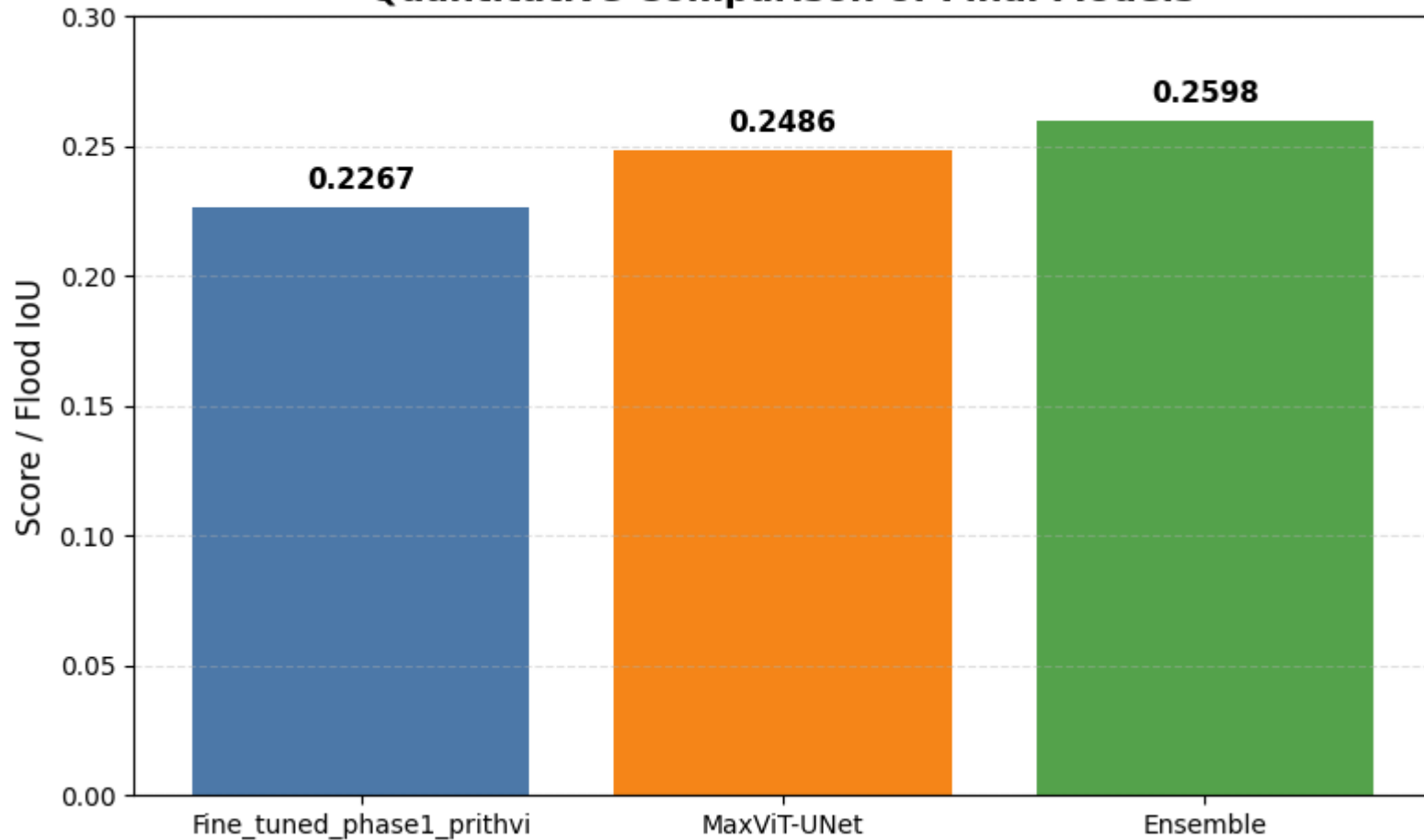
It works because the model is no longer relying on a single image snapshot; by adding pre-flood Sentinel-1 temporal change, water-history layers, and topographic context, it can better distinguish new floodwater from existing water bodies, which is the main challenge of Phase 2.

S4

We used a fine-tuned Prithvi-300M model and MaxViT-UNet, trained on GPU (Kaggle T4/P100) with full runs taking ~1–2 hours and inference taking a few seconds per 512×512 patch.

The main shift from Phase 1 to Phase 2 was moving from binary flood detection to 3-class flood/water/no-flood segmentation, enabled by adding May 10 Sentinel-1 temporal data, GSW layers, and DEM features to distinguish flood from permanent water.

Quantitative Comparison of Final Models



Details of Relevant Experimentation and Methodology

Data & Feature Engineering

Started from the Phase 1 flood dataset and extended it for Phase 2 three-class segmentation: no-flood, flood, water body.

Built an 18-band input stack by combining:

EOS-4 SAR (HH, HV)

Optical bands (Green, Red, NIR, SWIR)

Water index (MNDWI)

May 10 Sentinel-1 pre-flood SAR temporal features

JRC Global Surface Water layers (occurrence, recurrence, seasonality)

DEM-based features (elevation, slope, TWI)

Model Development

Fine-tuned IBM Prithvi-300M foundation model from Phase 1 binary flood detection to Phase 2 three-class segmentation.

In parallel, trained an independent MaxViT-UNet segmentation model for architectural diversity.

Final predictions were generated using ensemble inference from both model families.

Training Strategy

Used class-weighted loss to address flood-class imbalance and emphasize flood recall.

Applied data augmentation (flips, rotations, spatial transforms) and oversampling of flood-rich patches.

Performed threshold tuning and validation experiments on geographically closer patches (IDs 70–79) to better approximate the test distribution.

Key Experimentation Insights

The biggest gains came from:

Adding pre-flood Sentinel-1 temporal context

Using GSW + DEM auxiliary features

Combining Prithvi + MaxViT through ensembling

This shifted the approach from binary flood detection in Phase 1 to context-aware flood-vs-water-body discrimination in Phase 2.

External / Auxiliary Datasets Used

1) Sentinel-1 pre-flood SAR (May 10, 2024)

Purpose: Temporal pre-flood baseline for change/log-ratio features

Source: ASF Alaska SAR Search

Link: <https://search.asf.alaska.edu/>

2) JRC Global Surface Water (GSW)

Purpose: Historical water occurrence, recurrence, and seasonality to distinguish permanent water from flood

Source: Global Surface Water Explorer / JRC

Download Link: <https://global-surface-water.appspot.com/download>

Explorer Link: <https://global-surface-water.appspot.com/>

3) DEM / Terrain Data

Purpose: Elevation, slope, and TWI for flood-prone terrain characterization

Typical source used for derivatives: SRTM/NASA DEM

Reference Link: <https://www.earthdata.nasa.gov/data/catalog/lpcloud-srtmgl1-003>

Thank You

Final Notes on the Presentation

- You are not bound to stick by this template but remember to have most of the particulars mentioned in this template in your final presentation. You have to **prioritise on what information would be pivotal** to prove your case and approach as there is a **hard time limit of 5 minutes for you to present**.
- You are recommended to be as concise and cogent with your explanations, proofs, results, experimentation details etc. Ideally the explanations should be backed up by experimentation and possibly literature and not mere AI slop.
- Make sure to highlight what really differentiates your approach from the rest and its novelty. Truly focus on that portion, condense and refine that content for the presentation.
- All the very best!